MLOps Lab Assignment 2

Tracking a Pre-trained Sentiment Model with MLflow

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Objective

The objective of this lab is to use MLflow to track parameters, metrics and artifacts for a pre-trained Hugging Face sentiment-analysis model, inspect and compare experiment runs in the MLflow UI, and identify the best-performing run. No model training is performed; the model is used only for inference.

1. Which run had the lowest latency, and why might that be?

run_batch8 had the lowest latency at 0.120 seconds, followed by run_batch4 at 0.133 seconds and run_batch1 at 0.226 seconds. A larger batch size allows the model to process multiple sentences in a single batch, reducing processing overhead per sentence. The first run may also take longer because of model warm-up overhead.

2. Did accuracy change between runs? Why or why not?

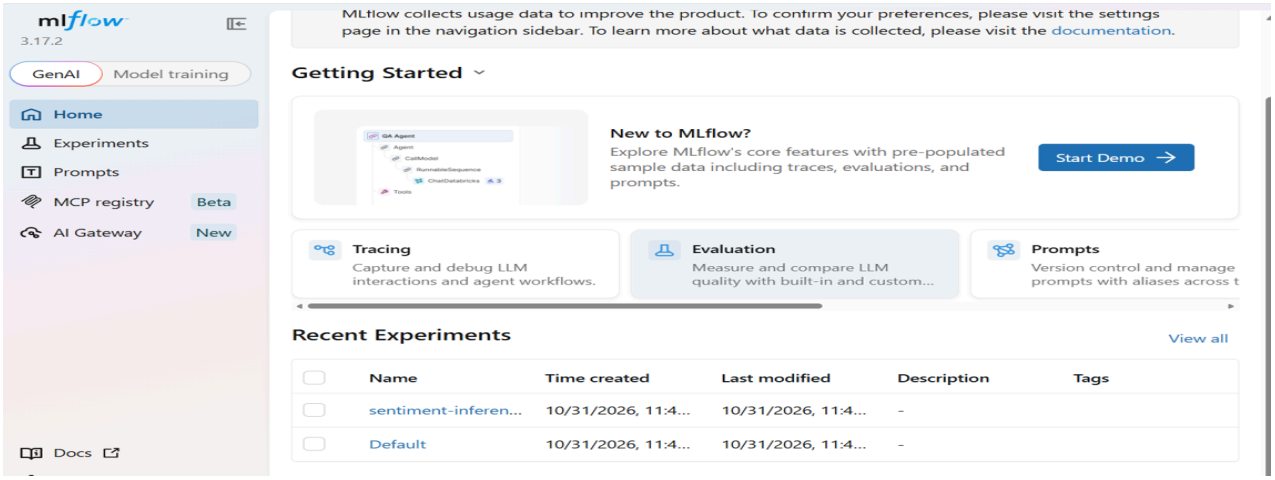
No, accuracy remained at 1.0 (100%) across all three runs. Each run used the same pre-trained sentiment analysis model and the same 10 test sentences. Since only the batch size changed, the predicted labels remained the same, so accuracy did not change.

3. Name two benefits of using MLflow here instead of printing results in the terminal.

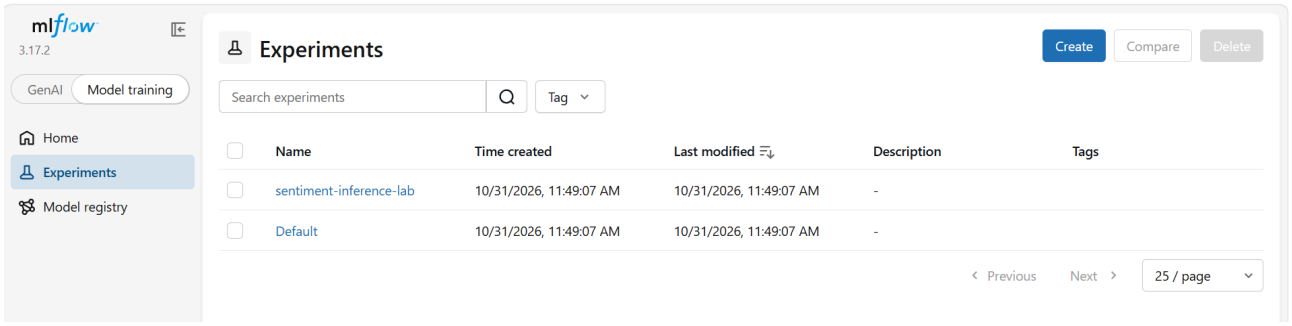
1. Reproducibility and record-keeping: MLflow logs the parameters, metrics, and prediction artifacts for each experiment run, making it easier to review results later.

2. Easy comparison: MLflow provides a dashboard to compare multiple runs side by side using tables and visualizations, making differences in latency, accuracy, and other metrics easier to understand than terminal output.

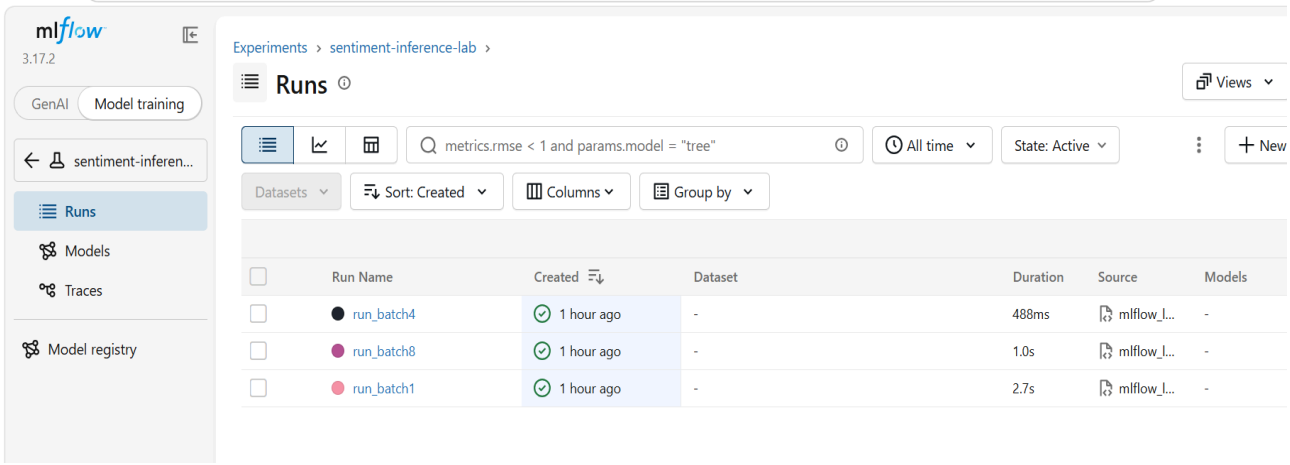
Screenshot 1: MLflow UI home page



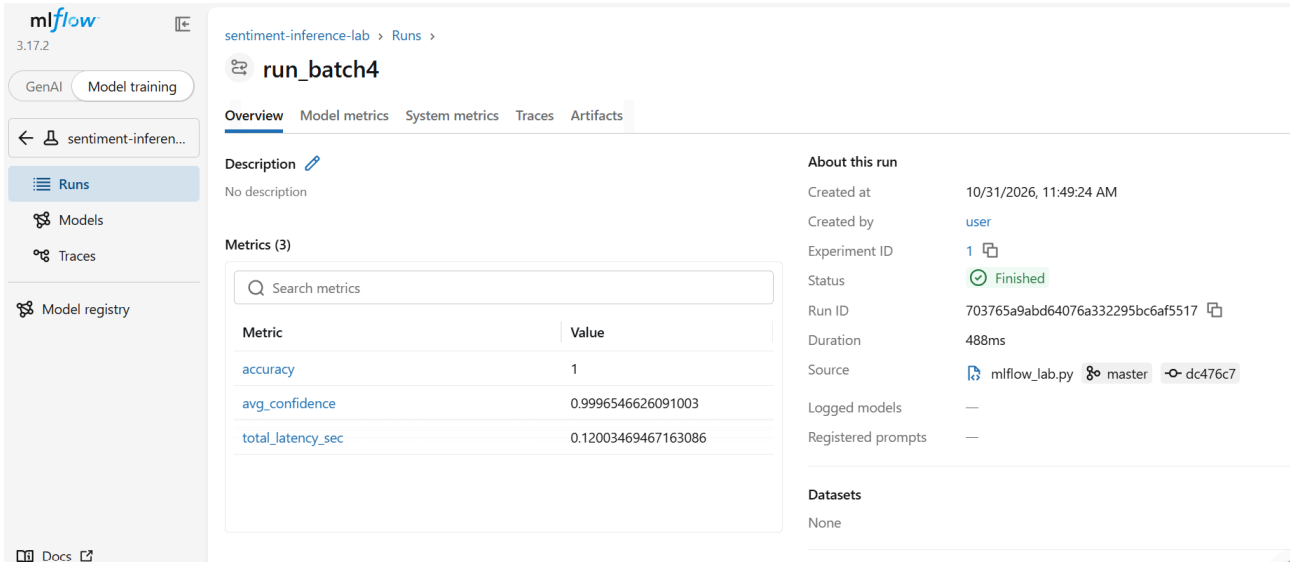
Screenshot 2: Experiments list

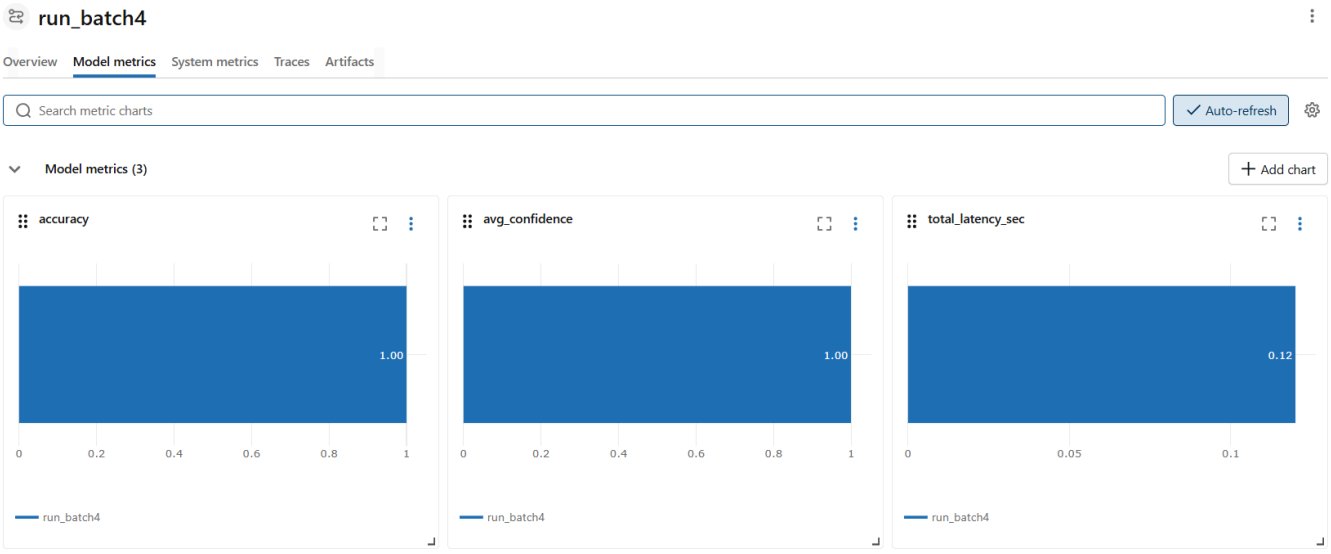


Screenshot 3: Runs in sentiment-inference-lab

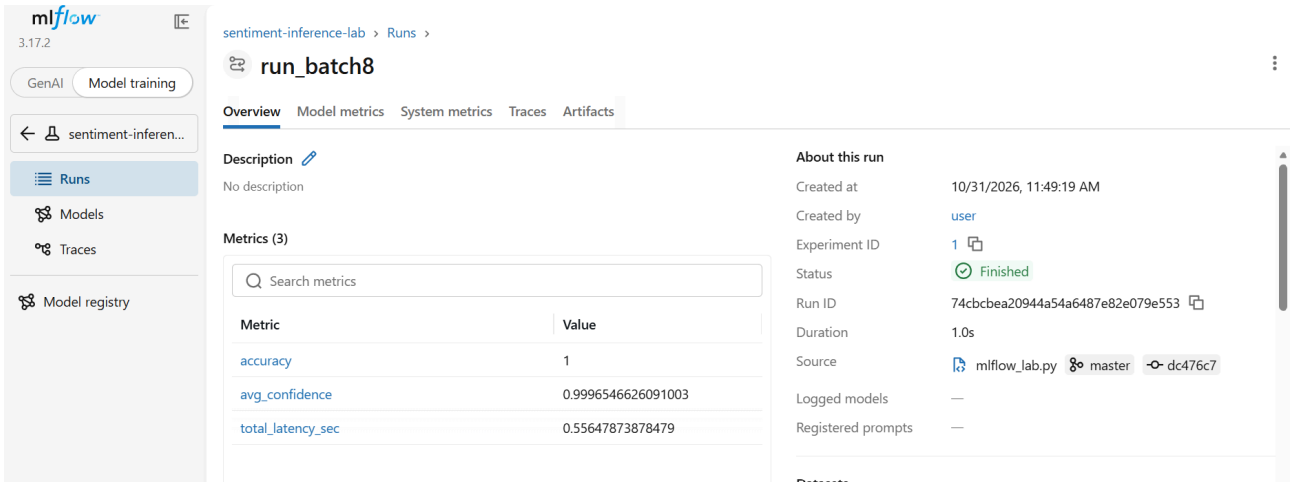


Screenshot 4: run_batch4 overview and metrics



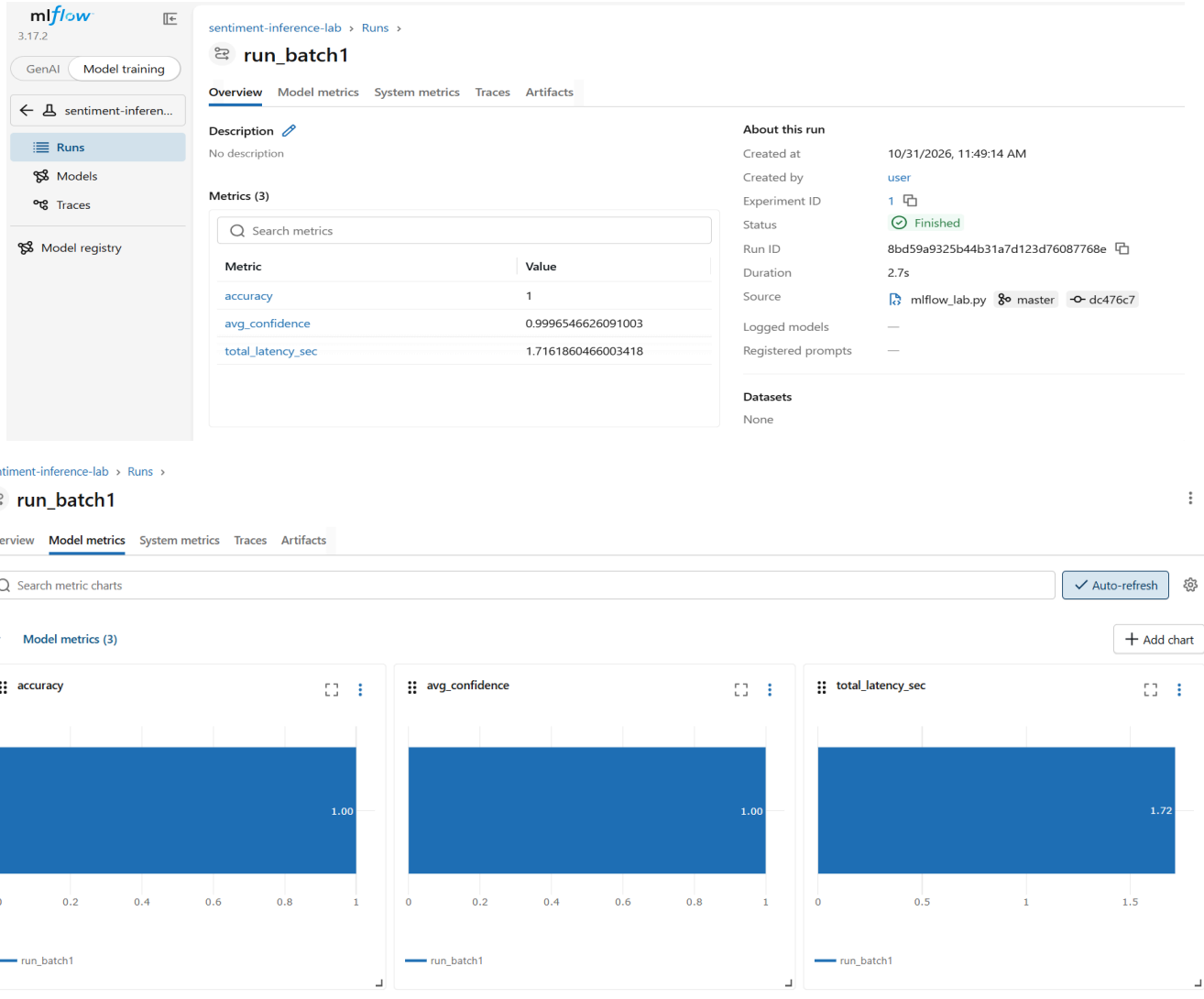


Screenshot 5: run_batch8 overview and metrics





Screenshot 6: run_batch1 overview and metrics



Screenshot 7: Comparing 3 Experiments

